



AHCI – Project Proposal

EasyMeds **(Elderly Medication Management App)**

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Introduction:

With the growing aging population, managing daily medication has become a critical challenge for elderly individuals. Many older adults are required to follow strict medication schedules involving multiple prescriptions, varying dosages, and specific timings. Failure to adhere to these schedules can lead to serious health complications.

This project proposes the design and development of a **mobile application tailored specifically for elderly users** to help them manage their medication routines effectively. The focus of this project is not only functionality but also **usability, accessibility, and human-centered design**, in line with Applied Human-Computer Interaction (HCI) principles.

Existing Interface Analysis and Improvements

This project improves on the existing interface named, **Medisafe**. While Medisafe is effective for general users, it assumes a level of digital literacy that many elderly users may not possess. It includes small text, complex navigation, and multiple features that may overwhelm users with limited technical experience.

Common issues observed in existing interface include:

- Overly cluttered screens with too much information
- Small buttons and low readability
- Use of medical or technical terminology
- Lack of clear feedback after user actions
- Complex setup processes

Proposed Improvements

The proposed application addresses these issues through the following design improvements:

- **Simplified Interface:** Minimalist design with only essential features visible
- **Improved Readability:** Large fonts and high-contrast color schemes
- **Easy Navigation:** Linear and predictable user flow
- **Clear Feedback:** Immediate confirmation after each action
- **Error Prevention:** Alerts to prevent duplicate or missed doses
- **User-Friendly Language:** Avoidance of technical or medical jargon
- **Caregiver Support:** Additional human support for verification of elderly patient activity

Functionalities:

The application will include the following features, each aligned with key HCI principles:

a. Medication Management

- Users can add, edit, and delete medications (name, dosage, timing, placement)
- Simple forms with minimal input fields to reduce cognitive load
- Use of familiar language instead of medical jargon

HCI Principle: Simplicity and match with real-world concepts

b. Reminder System

- Scheduled notifications reminding users when to take medication
- Audio reminders
- Persistent alerts until acknowledged

HCI Principle: Visibility of system status and timely feedback

c. Daily Checklist

- A clear list showing medications for the day
- Option to mark doses as “Taken” or “Missed”

HCI Principle: Recognition rather than recall (users don’t need to remember tasks)

d. Accessibility Features

- Large font sizes and scalable UI
- High contrast color schemes
- Simple layout with large touch targets

HCI Principle: Accessibility and usability for diverse user capabilities

e. Error Prevention

- Warning if a user tries to log a dose twice
- Alerts for missed doses with guidance

HCI Principle: Error prevention and recovery

f. Feedback System

- Visual and auditory confirmation after marking a dose
- Friendly messages to reassure users

HCI Principle: Immediate and clear feedback

g. Caregiver Support

- Caregiver notification system for missed doses
- Caregiver access to medical information of the elderly patient
- Emergency call button

HCI Principle: Flexibility and user support

Conclusion:

This project aims to design an accessible and user-friendly medication management application specifically for elderly users. By focusing on HCI principles such as simplicity, visibility, feedback, and error prevention, the app seeks to reduce common barriers faced by older adults when using digital technology.

The proposed solution not only addresses a real-world problem but also demonstrates the importance of **human-centered design in creating effective and inclusive interfaces**. This project highlights how thoughtful design can significantly improve usability and user experience for a vulnerable user group.